# PRAJWAL A

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## PROFESSIONAL SUMMARY

Software engineering graduate with hands-on experience building, deploying, and maintaining production software systems through a full-stack development internship. Developed and maintained two live production platforms, working across backend development, database design, API development, deployment, production support, and system improvements. Also built an AI-enabled capstone project integrating computer vision, embedded systems, and modern web technologies to solve real-world engineering problems. Interested in building reliable software, understanding system design, and solving practical engineering problems through continuous learning.

## EDUCATION

**B.Tech, Electronics and Communication Engineering** 2026
Dayananda Sagar University, Bengaluru, Karnataka

## PROFESSIONAL EXPERIENCE

**Full Stack Developer Intern** Feb 2026 – Present
*PulseWriteX Solutions Pvt. Ltd. — Bengaluru*

- Developed, deployed, and maintained production software throughout the full development lifecycle, from requirement analysis and backend development to deployment and production support.
- Worked across backend development, frontend implementation, database design, API development, and system maintenance while contributing to live production platforms.
- Collaborated with stakeholders to translate business requirements into reliable software solutions and continuously improve production systems.

## PROJECTS

### PulseX – Prototyping Services Platform

*React • TypeScript • Node.js • Express.js • MySQL • Prisma • Razorpay*

- Designed and developed a full-stack web platform that digitized quotation and order management for PCB manufacturing, 3D printing, laser cutting, and prototyping services.
- Engineered the backend with REST APIs, relational data models, authentication, and secure payment workflows; built a custom Gerber viewer rendering PCB designs at 0.1mm accuracy.
- Built responsive user and admin interfaces; deployed and maintain the platform solo (9+ months in production).

### Rizzo Education – Production Systems Engineering

*Node.js • PostgreSQL • Prisma • Linux • Nginx • PM2 • GitHub Actions*

- Engineered production improvements for a live AI education platform: strengthened application security, improved backend reliability, and resolved critical issues.
- Increased platform stability by redesigning workflows, and built a safer release process through migrations, automated backups, and a gated CI/CD pipeline.

### Omni Room Management System – AI Smart Classroom

*Python • OpenCV • Raspberry Pi • React • Node.js • Ollama • Computer Vision*

- Designed a smart classroom platform integrating embedded hardware, computer vision, and web technologies, including custom PTZ camera hardware on Raspberry Pi 4.
- Built a real-time face recognition pipeline (HOG + CNN after benchmarking five models) and integrated a local Ollama LLM that turns class audio into summaries and study aids.

### Fluid AI – Autonomous Agent

*Python • FastAPI • Groq LLM • Pydantic • python-docx*

- Built a four-stage autonomous pipeline (validate → plan → generate → build) that turns a natural-language brief into a formatted Word document.
- Enforced Pydantic validation with a retry loop that feeds validation errors back to the LLM.

### Predictive Maintenance – IIoT

*Python • Scikit-learn • Pandas • Edge ML*

- Built the preprocessing pipeline and benchmarked classical models for motor vibration and temperature anomaly detection, reaching 0.88 precision.
- Ran inference locally on edge hardware with no cloud dependency.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL, Java, C
**Frontend:** React, HTML5, CSS3
**Backend:** Node.js, Express.js | Databases: MySQL, PostgreSQL, Prisma ORM
**Production Systems:** Linux, Nginx, PM2, Git, GitHub Actions
**Computer Vision:** OpenCV, Ollama, Raspberry Pi